

Electric Vehicle Adoption Analysis and Range Prediction

Project Overview

This project analyzes Electric Vehicle population data and builds a machine learning model to predict electric vehicle range.

Dataset Information

Dataset Size: 279,780 rows and 16 columns containing manufacturer, model, vehicle type, electric range, and location details.

Task 1: Dataset Understanding

Explored dataset structure, size, features, data types, missing values, and duplicates.

Task 2: Data Cleaning

Handled missing values and prepared the dataset for analysis.

Task 3: Exploratory Data Analysis

Analyzed EV types, manufacturers, adoption trends, electric range distribution, and geographic patterns.

Task 4: Data Visualization

Created bar charts, line charts, pie charts, heatmaps, and box plots.

Task 5: Predictive Modeling

Random Forest Regressor achieved MAE=2.05, RMSE=9.47, and $R^2=0.9854$.

Key Findings

Tesla dominates registrations, BEVs are the majority, EV adoption surged after 2020, and King County leads EV adoption.

Conclusion

The project successfully demonstrates the complete data science workflow and highlights the growth of electric mobility.